10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Review Exercise 4 — Solved

Unit 4 | MCQ + Review Exercise | New Book (PECTAA)

Review Exercise 4 (Solved)

This exercise mixes the whole unit — functions, composition of functions, inverse functions, absolute value graphs, absolute value inequalities and real-life models. Each answer is written the way it should appear in the note-book.

Multiple Choice Questions

(i) If f(x) = (5x − 6)/3, then what is the value of f(3)?

(a) −1 (b) 3 (c) 9 (d) 15

Answer: (b) 3 — f(3) = (5 × 3 − 6)/3 = 9/3 = 3

(ii) A function f from X to Y is represented by:

(a) f: X × Y (b) f: Y → X (c) f: X → Y (d) f: X/Y

Answer: (c) f: X → Y — the arrow is drawn from the domain X towards the range Y

(iii) (fog)(x) =

(a) (f + g)(x) (b) (f − g)(x) (c) f(g(x)) (d) f(x) ÷ g(x)

Answer: (c) f(g(x)) — fog means f of g, i.e. put g(x) inside f

(iv) If f(x) = 2x + 3 and g(x) = x + 1, then f(x) + g(x) =

(a) 3x (b) 3x + 4 (c) 4 (d) $2x^2 + 3$

Answer: (b) 3x + 4 — 2x + 3 + x + 1 = 3x + 4

(v) If f(x) = 5x + 2 and h(x) = 2x − 2, then f(x) − h(x) =

(a) 3x (b) $5x^2 - 4$ (c) 3x + 4 (d) −3x − 4

Answer: (c) 3x + 4 — 5x + 2 − (2x − 2) = 5x + 2 − 2x + 2 = 3x + 4

(vi) If f(x) = 3x + 1 and g(x) = 2x, then g(x) × f(x) =

(a) 6x + 2x (b) $5x^2 + 1$ (c) x + 1 (d) $6x^2 + 2x$

Answer: (d) $6x^2 + 2x$ — $2x(3x + 1) = 6x^2 + 2x$

(vii) If $f(x) = x^2 - 4$, g(x) = x + 2, x ≠ −2, then f(x)/g(x) =

(a) 1/(x − 2) (b) 1/(x + 2) (c) x + 2 (d) x − 2

Answer: (d) x − 2 — $(x^2 - 4)/(x + 2) = (x - 2)(x + 2)/(x + 2) = x - 2$

(viii) What is the shape of the graph of an absolute value function?

(a) U-shaped (b) V-shaped (c) L-shaped (d) M-shaped

Answer: (b) V-shaped — |x| turns back at the vertex, so the two arms make a V

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(ix) If a graph represents a function, then every vertical line must intersect it at:

(a) 4 points (b) 3 points (c) 2 points (d) 1 point

Answer: (d) 1 point — this is the vertical line test — one input gives only one output

(x) If $f(x) = x^3$, then $f(-2) =$

(a) -8 (b) 8 (c) 4 (d) -6

Answer: (a) -8 — $f(-2) = (-2)^3 = -8$

2. If $f(x) = 25 - x^2$ and $g(x) = 5 + x$, then find

(i) $f(x) + g(x)$ (ii) $f(x) - g(x)$ (iii) $f(x)\cdot g(x)$

(iv) $f(x)/g(x)$ (v) $f(7)$ (vi) $g(-8)$

(i) $f(x) + g(x)$

$f(x) + g(x) = (25 - x^2) + (5 + x)$

$= 25 + 5 - x^2 + x$

Answer: $f(x) + g(x) = -x^2 + x + 30$

(ii) $f(x) - g(x)$

$f(x) - g(x) = (25 - x^2) - (5 + x)$

$= 25 - x^2 - 5 - x$

Answer: $f(x) - g(x) = -x^2 - x + 20$

(iii) $f(x)\cdot g(x)$

$f(x)\cdot g(x) = (25 - x^2)(5 + x)$

$= 125 + 25x - 5x^2 - x^3$

Answer: $f(x)\cdot g(x) = -x^3 - 5x^2 + 25x + 125$

(iv) $f(x)/g(x)$

$$\frac{25 - x^2}{5 + x} = \frac{(5 - x)(5 + x)}{5 + x}$$

Cancel the common factor $(5 + x)$

Answer: $f(x)/g(x) = 5 - x,\ x \neq -5$

(v) $f(7)$

$f(7) = 25 - (7)^2$

$= 25 - 49$

Answer: $f(7) = -24$

(vi) $g(-8)$

$g(-8) = 5 + (-8)$

Answer: $g(-8) = -3$

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3. If $f(x) = x^3$ and $g(x) = 14 + 2x$, then find

(i) (fog)(x) (ii) (gof)(x) (iii) (fof)(x) (iv) (gog)(x)

In (fog)(x) the inner function is g, so replace every x of f by g(x). In (gof)(x) it is the other way round.

(i) (fog)(x)

$(fog)(x) = f(g(x))$

$= f(14 + 2x)$

Answer: $(fog)(x) = (14 + 2x)^3$

(ii) (gof)(x)

$(gof)(x) = g(f(x))$

$= g(x^3)$

Answer: $(gof)(x) = 14 + 2x^3$

(iii) (fof)(x)

$(fof)(x) = f(f(x))$

$= f(x^3)$

$= (x^3)^3$

Answer: $(fof)(x) = x^9$

(iv) (gog)(x)

$(gog)(x) = g(g(x))$

$= g(14 + 2x)$

$= 14 + 2(14 + 2x)$

$= 14 + 28 + 4x$

Answer: $(gog)(x) = 4x + 42$

4. Find $f^{-1}(x)$, if

(i) $f(x) = 9x - 1$ (ii) $f(x) = 5/(x - 1),\ x \neq 1$

(iii) $f(x) = \sqrt{(x - 5)},\ x \geq 5$ (iv) $f(x) = (3 - x)/2$

Method: write $y = f(x)$, make x the subject, and in the last step replace y by x. The answer is $f^{-1}(x)$.

(i) $f(x) = 9x - 1$

Let $y = 9x - 1$

$9x = y + 1$

$x = \frac{y + 1}{9}$

Replace y by x

Answer: $f^{-1}(x) = (x + 1)/9$

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(ii) $f(x) = 5/(x - 1), x \neq 1$

Let $y = \frac{5}{x - 1}$

$y(x - 1) = 5$

$x - 1 = \frac{5}{y}$

$x = 1 + \frac{5}{y}$

$x = \frac{y + 5}{y}$

Replace y by x

Answer: $f^{-1}(x) = (x + 5)/x, x \neq 0$

(iii) $f(x) = \sqrt{(x - 5)}, x \geq 5$

Let $y = \sqrt{x - 5}$

Square both sides

$y^2 = x - 5$

$x = y^2 + 5$

Replace y by x

$f^{-1}(x) = x^2 + 5$

The range of f is $y \geq 0$, so the domain of f^{-1} is $x \geq 0$

Answer: $f^{-1}(x) = x^2 + 5, x \geq 0$

(iv) $f(x) = (3 - x)/2$

Let $y = \frac{3 - x}{2}$

$2y = 3 - x$

$x = 3 - 2y$

Replace y by x

Answer: $f^{-1}(x) = 3 - 2x$

5. Plot graph for the following absolute valued functions.

(i) $f(x) = 7|x|$ (ii) $f(x) = |x + 6| - 2$

(i) $f(x) = 7|x|$

This is of the form $f(x) = a|x - h| + k$

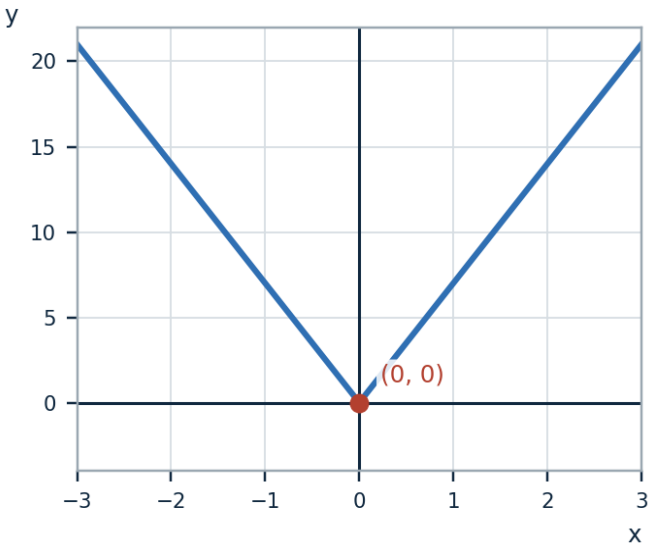
Here $a = 7, h = 0, k = 0 \implies$ vertex (0, 0)

Table of values

f(x)

x	−2	−1	0	1	2
y	14	7	0	7	14

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The V-shaped graph has its vertex at (0, 0)

It is symmetric about the line x = 0

Answer: f(x) = 7|x| has vertex (0, 0)

(ii) f(x) = |x + 6| − 2

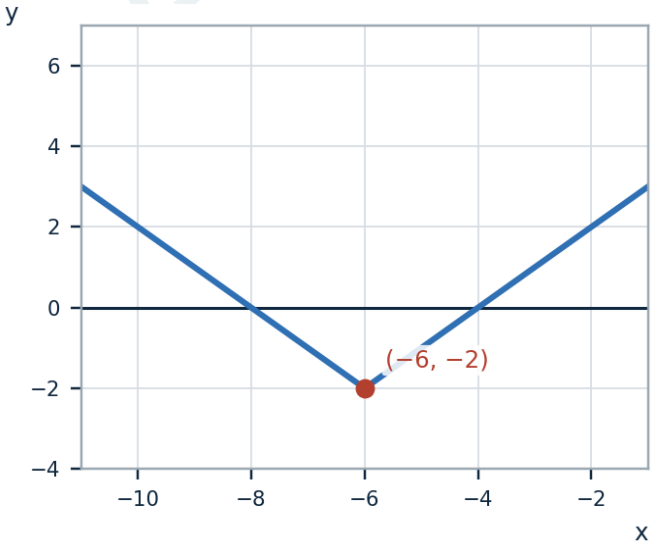
This is of the form f(x) = a|x − h| + k

Here a = 1, h = −6, k = −2 ⟹ vertex (−6, −2)

Table of values

f(x)

x	−9	−8	−6	−4	−3
y	1	0	−2	0	1



The V-shaped graph has its vertex at (−6, −2)

It is symmetric about the line x = −6

Answer: f(x) = |x + 6| − 2 has vertex (−6, −2)

6. Solve and express the solution on number line.

(i) $|3x - 2| = 1$ (ii) $|6x + 1| = 9$

When $|X| = a$, we write $X = a$ or $X = -a$, and solve both equations. The two answers are shown by closed dots on the number line.

(i) $|3x - 2| = 1$

$3x - 2 = 1$ or $3x - 2 = -1$

$3x = 3$ or $3x = 1$

$x = 1$ or $x = \frac{1}{3}$

Solution set = {1, 1/3}



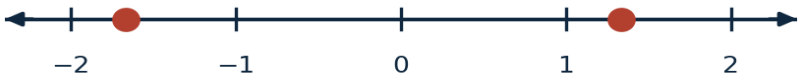
Answer: x = 1 or x = 1/3

(ii) $|6x + 1| = 9$

$6x + 1 = 9$ or $6x + 1 = -9$

$6x = 8$ or $6x = -10$

$x = \frac{4}{3}$ or $x = -\frac{5}{3}$



Answer: x = 4/3 or x = −5/3

7. Solve and express the solution on number line.

(i) $|7 - 2x| \leq 1$ (ii) $|6x + 18| \leq 24$

$|X| \leq a$ means $-a \leq X \leq a$. Solve it as one double inequality, and remember to reverse the signs when you divide by a negative number.

(i) $|7 - 2x| \leq 1$

$-1 \leq 7 - 2x \leq 1$

Subtract 7 from every part

$-8 \leq -2x \leq -6$

Divide every part by -2 and reverse the signs

$4 \geq x \geq 3$

$3 \leq x \leq 4$



Answer: $3 \leq x \leq 4$, i.e. $x \in [3, 4]$

(ii) $|6x + 18| \leq 24$

$-24 \leq 6x + 18 \leq 24$

Subtract 18 from every part

$-42 \leq 6x \leq 6$

Divide every part by 6

$-7 \leq x \leq 1$



Answer: $-7 \leq x \leq 1$, i.e. $x \in [-7, 1]$

8. Given that $f(x) = 3x + 7$, find

(i) $f^{-1}(x)$

(ii) the value of x for which $f(x) = f^{-1}(x)$

(i) $f^{-1}(x)$

Let $y = 3x + 7$

$3x = y - 7$

$x = \frac{y - 7}{3}$

Replace y by x

Answer: $f^{-1}(x) = (x - 7)/3$

(ii) x for which $f(x) = f^{-1}(x)$

$3x + 7 = \frac{x - 7}{3}$

Multiply both sides by 3

$9x + 21 = x - 7$

$9x - x = -7 - 21$

$8x = -28$

$x = -\frac{28}{8} = -3.5$

Answer: x = −3.5, i.e. x = −7/2

9. A company earns a profit of Rs. 100 for each item sold after a fixed monthly expense of Rs. 10,000. A function P(x) = 100x − 10,000 represents the profit after selling x items. How many items must be sold to break even (profit = 0)?

Solution

For break even, put P(x) = 0

$100x - 10,000 = 0$

$100x = 10,000$

$x = \frac{10000}{100}$

$x = 100$

Answer: The company must sell 100 items to break even

10. A store offers a 15% discount on a product. A function D(p) = 0.85p gives the selling price after the discount on an original price p. What is the selling price of an item that originally costs Rs. 2,000?

Solution

D(p) = 0.85p

Put p = 2000

$D(2000) = 0.85 \times 2000$

Answer: Selling price = Rs. 1,700

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11. A GPS system is considered accurate if the actual position differs from the reported position by no more than 6 metres. If the actual location is at point x = 100, the allowed range is defined as $|x - r| \leq 6$, where r is the reported location. What is the range of acceptable reported locations?

Solution

$|x - r| \leq 6$

Put $x = 100$

$|100 - r| \leq 6$

$-6 \leq 100 - r \leq 6$

Subtract 100 from every part

$-106 \leq -r \leq -94$

Multiply every part by -1 and reverse the signs

$106 \geq r \geq 94$

$94 \leq r \leq 106$



Answer: The reported location must satisfy $94 \leq r \leq 106$ metres

Remember — Unit 4 at a glance

1. A function gives exactly one output for every input; it passes the vertical line test.
2. $(fog)(x) = f(g(x))$ — the inner function g is done first.
3. To find $f^{-1}(x)$: write $y = f(x)$, make x the subject, then replace y by x. Also swap the domain and the range.
4. $y = a|x - h| + k$ is V-shaped with vertex (h, k), symmetric about $x = h$.
5. $|X| = a \Longrightarrow X = a$ or $X = -a$; $|X| < a \Longrightarrow -a < X < a$; $|X| > a \Longrightarrow X < -a$ or $X > a$.